

Database Schema Reference

Every table, view, column, constraint, index, trigger, function and role of the catalog database.

Project	Digital Twin Catalog (NARAGA 3D Catalog), reusable 3DCityDB template
Document	Database Schema
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Contents

Introduction

Conventions

Schemas

Schema catalog

catalog.v_layer

catalog.dataset

catalog.import_job

catalog.import_job_layer

catalog.layer

catalog.ref_generated_by

catalog.ref_layer_status

catalog.ref_lod_level

catalog.ref_theme

catalog.ref_tileset_status

catalog.tileset

catalog.layer_feature_stats

Schema audit

audit.change_log

Functions

Roles and privileges

Introduction

This document is the reference for every table, view, column, constraint, index, trigger, function and role added by the catalog migrations. It is generated from the live database catalogue (PostgreSQL 16.4, PostGIS 3.4.3, 3DCityDB 5.1.4). The 3DCityDB schema `citydb` is documented by its project and is not repeated here.

Conventions

Topic	Convention
Migrations	Plain SQL files in <code>db/migrations</code> , applied by dbmate in order. Applied files are never edited.
Keys	<code>uuid</code> primary keys (<code>gen_random_uuid()</code>); reference tables use a short text <code>code</code> .
Names and time	<code>snake_case</code> names, <code>timestamp_tz</code> for every time.
Coordinates	<code>bbox</code> is <code>geometry(Polygon, 4326)</code> , computed from the data, never typed. The source CRS is only recorded in <code>dataset.crs_epsg</code> ; the database SRID is fixed at creation.
LOD	0 to 4 (CityGML 2.0). LOD4 of CityGML 2.0 is stored by 3DCityDB as deprecated properties; layers still use <code>lod = 4</code> .
Lineage tag	<code>dataset.theme.lodN</code> in <code>citydb.feature.lineage</code> links 3D objects to a layer.
Code lists	Reference tables, not ENUM types. Add a row to extend; no migration needed.
Audit columns	<code>created_at</code> , <code>created_by</code> , <code>updated_at</code> , <code>updated_by</code> on <code>dataset</code> , <code>import_job</code> , <code>layer</code> , <code>tileset</code> . Filled by trigger from the signed-in user.
Mandatory user	Set <code>app.user</code> per transaction: <code>SELECT set_config('app.user', 'name', true);</code> . Without it every write is rejected.
Deletion	Archive through status. DELETE is allowed to <code>twin_admin</code> only. Old rows stay in <code>audit.change_log</code> .
Comments	Every table and column has a plain-language <code>COMMENT</code> , shown in the admin UI.

Schemas

Schema	Purpose	Tables	Views
catalog	Reference lists, datasets, layers, tilesets and import history. Editable by trained staff through the admin UI.	10	2
audit	Automatic history of every change to catalog data.	1	0

Other schemas: `citydb` (3DCityDB v5, untouched except the index `feature_lineage_inx`) and `app` (Django tables).

Schema catalog

Reference lists, datasets, layers, tilesets and import history. Editable by trained staff through the admin UI.

catalog.v_layer

view. One row per layer with its active tileset, object count and stale flag. Used by the API and the inventory.

Column	Type	Null	Default	Meaning
layer_id	uuid	yes	-	-
dataset_id	uuid	yes	-	-
dataset_code	text	yes	-	-
dataset_name	text	yes	-	-
generated_by	text	yes	-	-
attribution	text	yes	-	-
vertical_datum	text	yes	-	-
region	text	yes	-	-
theme_code	text	yes	-	-
lod	smallint	yes	-	-
title	text	yes	-	-
status	text	yes	-	-
bbox_minx	double precision	yes	-	-
bbox_miny	double precision	yes	-	-
bbox_maxx	double precision	yes	-	-
bbox_maxy	double precision	yes	-	-
bbox	geometry(Polygon,4326)	yes	-	-
feature_count	bigint	yes	-	-
tileset_id	uuid	yes	-	-
tileset_version	integer	yes	-	-
provider	text	yes	-	-
ion_asset_id	bigint	yes	-	-
url	text	yes	-	-
height_offset_m	numeric	yes	-	-

Column	Type	Null	Default	Meaning
published_at	timestamp with time zone	yes	-	-
source_snapshot_at	timestamp with time zone	yes	-	-
is_stale	boolean	yes	-	-
updated_at	timestamp with time zone	yes	-	-
updated_by	text	yes	-	-

catalog.dataset

table. A collection of data for one area from one source, e.g. OIKN.

Column	Type	Null	Default	Meaning
id	uuid	no	gen_random_uuid()	-
code	text	no	-	Short unique name (a-z, 0-9, _). It is the tag stored on every 3D object, so it cannot be changed once data is imported.
name	text	no	-	-
description	text	yes	-	-
region	text	yes	-	-
source_system	text	yes	-	-
owner_org	text	yes	-	-
contact_email	text	yes	-	-
generated_by	text	no	'external'::text	-
crs_epsg	integer	yes	-	Coordinate system of the SOURCE files (record only). The database itself has one fixed coordinate system.
vertical_datum	text	no	'unknown'::text	Height reference of the source: ellipsoid, egm2008, egm96, local or unknown. Needed to place tiles correctly in Cesium.
license	text	yes	-	-

Column	Type	Null	Default	Meaning
attribution	text	yes	-	Credit line shown on the map (copyright/source).
bbox	geometry(Polygon,4326)	yes	-	Computed automatically from the data after import. Do not type by hand.
status	text	no	'active'::text	active or archived. Archiving a dataset archives all its layers.
created_at	timestamp with time zone	no	now()	-
created_by	text	no	' '::text	Filled automatically with the signed-in user.
updated_at	timestamp with time zone	no	now()	-
updated_by	text	no	' '::text	-

Constraints:

- `CHECK ((code ~ '^[a-z0-9_]+$'::text))`
- `CHECK ((vertical_datum = ANY (ARRAY['ellipsoid'::text, 'egm2008'::text, 'egm96'::text, 'local'::text, 'unknown'::text])))`
- `CHECK ((status = ANY (ARRAY['active'::text, 'archived'::text])))`
- `PRIMARY KEY (id)`
- `UNIQUE (code)`
- `FOREIGN KEY (generated_by) REFERENCES catalog.ref_generated_by(code)`

Indexes:

- `CREATE UNIQUE INDEX dataset_code_key USING btree (code)`
- `CREATE INDEX dataset_bbox_gix USING gist (bbox)`

Triggers:

- `a_stamp`: BEFORE INSERT OR UPDATE ON catalog.dataset FOR EACH ROW runs `catalog.stamp()`
- `z_audit`: AFTER INSERT OR DELETE OR UPDATE ON catalog.dataset FOR EACH ROW runs `audit.log_change('id')`
- `b_guard`: BEFORE UPDATE OF code ON catalog.dataset FOR EACH ROW runs `catalog.dataset_guard()`
- `c_archive`: AFTER UPDATE OF status ON catalog.dataset FOR EACH ROW runs `catalog.dataset_archive_cascade()`

catalog.import_job

table. One uploaded/imported file and what happened to it (the upload registry).

Column	Type	Null	Default	Meaning
id	uuid	no	gen_random_uuid()	-
dataset_id	uuid	no	-	-
tool	text	no	'citydb-tool'::text	-
tool_version	text	yes	-	-
citygml_version	text	yes	-	-
source_file	text	no	-	-
source_sha256	text	yes	-	Fingerprint of the file, to prove which exact file was imported.
source_size_bytes	bigint	yes	-	-
import_mode	text	no	'import_all'::text	-
lineage_tag	text	yes	-	Tag written on imported objects: dataset.theme lodN.
status	text	no	'running'::text	-
validation_passed	boolean	yes	-	True when geometry/count checks (e.g. validity) passed. Required before a layer can be validated.
counts	jsonb	no	'{}'::jsonb	-
report	jsonb	no	'{}'::jsonb	Detailed validation report (JSON).
started_at	timestamp with time zone	no	now()	-
finished_at	timestamp with time zone	yes	-	-
created_at	timestamp with time zone	no	now()	-
created_by	text	no	' '::text	-

Column	Type	Null	Default	Meaning
updated_at	timestamp with time zone	no	now()	-
updated_by	text	no	' '::text	-

Constraints:

- `CHECK ((source_sha256 ~ '^[0-9a-f]{64}$'::text))`
- `CHECK ((import_mode = ANY (ARRAY['import_all'::text, 'skip'::text, 'delete'::text, 'terminate'::text])))`
- `CHECK ((status = ANY (ARRAY['running'::text, 'succeeded'::text, 'failed'::text])))`
- `PRIMARY KEY (id)`
- `FOREIGN KEY (dataset_id) REFERENCES catalog.dataset(id)`

Indexes:

- `CREATE INDEX import_job_dataset_ix USING btree (dataset_id, started_at DESC)`

Triggers:

- `a_stamp`: BEFORE INSERT OR UPDATE ON catalog.import_job FOR EACH ROW runs `catalog.stamp()`
- `z_audit`: AFTER INSERT OR DELETE OR UPDATE ON catalog.import_job FOR EACH ROW runs `audit.log_change('id')`
- `b_guard`: BEFORE INSERT OR UPDATE ON catalog.import_job FOR EACH ROW runs `catalog.import_job_guard()`

catalog.import_job_layer

table. Which layers a file filled (one file can fill several).

Column	Type	Null	Default	Meaning
job_id	uuid	no	-	-
layer_id	uuid	no	-	-

Constraints:

- `PRIMARY KEY (job_id, layer_id)`
- `FOREIGN KEY (job_id) REFERENCES catalog.import_job(id) ON DELETE CASCADE`
- `FOREIGN KEY (layer_id) REFERENCES catalog.layer(id)`

Triggers:

- `z_audit`: AFTER INSERT OR DELETE OR UPDATE ON catalog.import_job_layer FOR EACH ROW runs `audit.log_change('job_id', 'layer_id')`

catalog.layer

table. What the frontend chooses: one theme at one LOD of a dataset (e.g. OIKN buildings LOD1).

Column	Type	Null	Default	Meaning
id	uuid	no	gen_random_uuid()	-
dataset_id	uuid	no	-	-
theme_code	text	no	-	-
lod	smallint	no	-	-
title	text	yes	-	-
description	text	yes	-	-
status	text	no	'draft'::text	draft → validated → published → archived. Moves are checked by the database.
validated_by_job_id	uuid	yes	-	The successful import that justified validation.
bbox	geometry(Polygon,4326)	yes	-	Computed automatically after import.
created_at	timestamp with time zone	no	now()	-
created_by	text	no	' '::text	-
updated_at	timestamp with time zone	no	now()	-
updated_by	text	no	' '::text	-

Constraints:

- PRIMARY KEY (id)
- UNIQUE (dataset_id, theme_code, lod)
- FOREIGN KEY (dataset_id) REFERENCES catalog.dataset(id)
- FOREIGN KEY (theme_code) REFERENCES catalog.ref_theme(code)
- FOREIGN KEY (lod) REFERENCES catalog.ref_lod_level(lod)
- FOREIGN KEY (status) REFERENCES catalog.ref_layer_status(code)
- FOREIGN KEY (validated_by_job_id) REFERENCES catalog.import_job(id)

Indexes:

- CREATE UNIQUE INDEX layer_dataset_id_theme_code_lod_key USING btree (dataset_id, theme_code, lod)

- `CREATE INDEX layer_lod_status_ix USING btree (lod, status)`
- `CREATE INDEX layer_bbox_gix USING gist (bbox)`

Triggers:

- `a_stamp` : BEFORE INSERT OR UPDATE ON catalog.layer FOR EACH ROW runs `catalog.stamp()`
- `z_audit` : AFTER INSERT OR DELETE OR UPDATE ON catalog.layer FOR EACH ROW runs `audit.log_change('id')`
- `b_guard` : BEFORE UPDATE ON catalog.layer FOR EACH ROW runs `catalog.layer_guard()`
- `c_archive` : AFTER UPDATE OF status ON catalog.layer FOR EACH ROW runs `catalog.layer_archive_cascade()`
- `y_published_needs_tileset` : AFTER INSERT OR UPDATE ON catalog.layer DEFERRABLE INITIALLY DEFERRED FOR EACH ROW runs `catalog.check_published_has_tileset()`

catalog.ref_generated_by

table. Who produced the data: an external provider, NARAGA, or manual work.

Column	Type	Null	Default	Meaning
code	text	no	-	-
name	text	no	-	-
description	text	yes	-	-

Constraints:

- `CHECK ((code ~ '^[a-z0-9_]+$'::text))`
- `PRIMARY KEY (code)`

Triggers:

- `z_audit` : AFTER INSERT OR DELETE OR UPDATE ON catalog.ref_generated_by FOR EACH ROW runs `audit.log_change('code')`

catalog.ref_layer_status

table. Lifecycle of a layer. Allowed moves are enforced by the database.

Column	Type	Null	Default	Meaning
code	text	no	-	-
name	text	no	-	-
description	text	yes	-	-
sort	integer	no	-	-

Constraints:

- `PRIMARY KEY (code)`

Triggers:

- `z_audit` : AFTER INSERT OR DELETE OR UPDATE ON `catalog.ref_layer_status` FOR EACH ROW runs `audit.log_change('code')`

catalog.ref_lod_level

table. Level of Detail per CityGML 2.0 (0 = footprint/terrain ... 4 = interior).

Column	Type	Null	Default	Meaning
<code>lod</code>	<code>smallint</code>	no	-	-
<code>name</code>	<code>text</code>	no	-	-
<code>description</code>	<code>text</code>	yes	-	-

Constraints:

- `CHECK (((lod ≥ 0) AND (lod ≤ 4)))`
- `PRIMARY KEY (lod)`

Triggers:

- `z_audit` : AFTER INSERT OR DELETE OR UPDATE ON `catalog.ref_lod_level` FOR EACH ROW runs `audit.log_change('lod')`

catalog.ref_theme

table. Subject of a layer: building, bridge, road, terrain ... Add a row to support a new theme.

Column	Type	Null	Default	Meaning
<code>code</code>	<code>text</code>	no	-	Short lowercase code, used in the lineage tag (<code>dataset.theme.lodN</code>). Do not change after use.
<code>name</code>	<code>text</code>	no	-	-
<code>description</code>	<code>text</code>	yes	-	-
<code>citydb_objectclass_hint</code>	<code>text[]</code>	yes	-	Typical 3DCityDB class names for this theme (documentation only).
<code>sort</code>	<code>integer</code>	no	<code>100</code>	-

Constraints:

- `CHECK ((code ~ '^[a-z0-9_]+$' :: text))`
- `PRIMARY KEY (code)`

Triggers:

- `z_audit` : AFTER INSERT OR DELETE OR UPDATE ON catalog.ref_theme FOR EACH ROW runs `audit.log_change('code')`

catalog.ref_tileset_status

table. Lifecycle of one 3D Tiles build.

Column	Type	Null	Default	Meaning
<code>code</code>	<code>text</code>	no	-	-
<code>name</code>	<code>text</code>	no	-	-
<code>description</code>	<code>text</code>	yes	-	-
<code>sort</code>	<code>integer</code>	no	-	-

Constraints:

- `PRIMARY KEY (code)`

Triggers:

- `z_audit` : AFTER INSERT OR DELETE OR UPDATE ON catalog.ref_tileset_status FOR EACH ROW runs `audit.log_change('code')`

catalog.tileset

table. One 3D Tiles build of a layer. Only the active one is served.

Column	Type	Null	Default	Meaning
<code>id</code>	<code>uuid</code>	no	<code>gen_random_uuid()</code>	-
<code>layer_id</code>	<code>uuid</code>	no	-	-
<code>version</code>	<code>integer</code>	no	-	-
<code>provider</code>	<code>text</code>	no	-	-
<code>ion_asset_id</code>	<code>bigint</code>	yes	-	Cesium ion asset number (required when provider is cesium_ion).
<code>url</code>	<code>text</code>	yes	-	tileset.json address (required when provider is self_hosted).

Column	Type	Null	Default	Meaning
converter_name	text	yes	-	-
converter_version	text	yes	-	-
source_snapshot_at	timestamp with time zone	yes	-	Data state used for this build. If the data changed later, the layer shows as stale.
height_offset_m	numeric	no	0	Vertical shift (metres) applied so buildings sit on the terrain.
status	text	no	'building'::text	-
is_active	boolean	no	false	The one build currently served. Only a Ready tileset can be active.
published_at	timestamp with time zone	yes	-	-
created_at	timestamp with time zone	no	now()	-
created_by	text	no	' '::text	-
updated_at	timestamp with time zone	no	now()	-
updated_by	text	no	' '::text	-

Constraints:

- `CHECK ((version > 0))`
- `CHECK ((provider = ANY (ARRAY['cesium_ion'::text, 'self_hosted'::text])))`
- `CHECK (((provider = 'cesium_ion'::text) AND (ion_asset_id IS NOT NULL)) OR ((provider = 'self_hosted'::text) AND (url IS NOT NULL)))`
- `CHECK (((NOT is_active) OR (status = 'ready'::text)))`
- `PRIMARY KEY (id)`
- `UNIQUE (layer_id, version)`
- `EXCLUDE USING btree (layer_id WITH =) WHERE (is_active) DEFERRABLE INITIALLY DEFERRED`
- `FOREIGN KEY (layer_id) REFERENCES catalog.layer(id)`
- `FOREIGN KEY (status) REFERENCES catalog.ref_tileset_status(code)`

Indexes:

- `CREATE UNIQUE INDEX tileset_layer_id_version_key USING btree (layer_id, version)`
- `CREATE INDEX tileset_one_active_per_layer USING btree (layer_id) WHERE is_active`

Triggers:

- `a_stamp` : BEFORE INSERT OR UPDATE ON catalog.tileset FOR EACH ROW runs `catalog.stamp()`
- `z_audit` : AFTER INSERT OR DELETE OR UPDATE ON catalog.tileset FOR EACH ROW runs `audit.log_change('id')`
- `b_guard` : BEFORE INSERT OR UPDATE ON catalog.tileset FOR EACH ROW runs `catalog.tileset_guard()`
- `y_published_needs_tileset` : AFTER INSERT OR DELETE OR UPDATE ON catalog.tileset DEFERRABLE INITIALLY DEFERRED FOR EACH ROW runs `catalog.check_published_has_tileset()`

catalog.layer_feature_stats

materialized view. Number of 3D objects per dataset tag, class and LOD. Refreshed after each import.

Column	Type	Null	Default	Meaning
<code>lineage</code>	<code>text</code>	yes	-	-
<code>dataset_code</code>	<code>text</code>	yes	-	-
<code>theme_code</code>	<code>text</code>	yes	-	-
<code>classname</code>	<code>text</code>	yes	-	-
<code>is_toplevel</code>	<code>boolean</code>	yes	-	-
<code>lod</code>	<code>text</code>	yes	-	-
<code>feature_count</code>	<code>bigint</code>	yes	-	-

Indexes:

- `CREATE UNIQUE INDEX layer_feature_stats_uq USING btree (lineage, classname, lod)`

Schema audit

Automatic history of every change to catalog data.

audit.change_log

table. Automatic history of every change to catalog data: who, when, old and new values. Cannot be edited.

Column	Type	Null	Default	Meaning
id	bigint	no	nextval('audit.change_log_id_seq'::regclass)	-
at	timestamp with time zone	no	now()	-
app_user	text	no	-	-
db_user	text	no	CURRENT_USER	-
table_name	text	no	-	-
pk	text	no	-	-
op	text	no	-	-
old_row	jsonb	yes	-	-
new_row	jsonb	yes	-	-

Constraints:

- CHECK ((op = ANY (ARRAY['INSERT'::text, 'UPDATE'::text, 'DELETE'::text])))
- PRIMARY KEY (id)

Indexes:

- CREATE INDEX change_log_record_ix USING btree (table_name, pk, at DESC)
- CREATE INDEX change_log_user_ix USING btree (app_user, at DESC)
- CREATE INDEX change_log_at_ix USING btree (at DESC)

Functions

Function	Returns	Security	Purpose
<code>audit.current_app_user()</code>	text	invoker	Returns the signed-in user name for this transaction; raises an error when empty.
<code>audit.log_change()</code>	trigger	definer	Trigger: writes one row to <code>audit.change_log</code> with old and new values.
<code>catalog.check_published_has_tileset()</code>	trigger	invoker	Deferred constraint trigger: a published layer has exactly one active Ready tileset at COMMIT.
<code>catalog.dataset_archive_cascade()</code>	trigger	invoker	Trigger: archiving a dataset archives its layers.
<code>catalog.dataset_guard()</code>	trigger	definer	Trigger: blocks changing <code>dataset.code</code> once 3D objects carry the lineage tag.
<code>catalog.import_job_guard()</code>	trigger	invoker	Trigger: sets <code>finished_at</code> when a job leaves the running state.
<code>catalog.layer_archive_cascade()</code>	trigger	invoker	Trigger: archiving a layer retires its tilesets.
<code>catalog.layer_guard()</code>	trigger	invoker	Trigger: layer identity freeze, allowed status moves, validation and publish preconditions.
<code>catalog.layer_lineage(dataset_code text, theme_code text, lod smallint)</code>	text	invoker	Builds the lineage tag <code>dataset.theme.lodN</code> for a layer.
<code>catalog.refresh_derived()</code>	void	definer	Run after an import: refreshes object counts and recomputes bounding boxes.
<code>catalog.stamp()</code>	trigger	invoker	Trigger: fills <code>created_by/updated_by</code> and timestamps from the signed-in user.
<code>catalog.tileset_guard()</code>	trigger	invoker	Trigger: allowed tileset status moves and <code>published_at</code> .

Roles and privileges

Roles are NOLOGIN group roles. Create one login user per person or service and grant one group, for example `CREATE ROLE api LOGIN PASSWORD '...' IN ROLE twin_ro;`.

Role	Login	Member of	Purpose
twin_admin	no	twin_editor	Editor rights plus DELETE and refresh_derived()
twin_editor	no	twin_ro	Insert and update catalog rows (admin UI)
twin_importer	no	twin_editor	Editor rights plus writing citydb (citydb-tool) and refresh_derived()
twin_ro	no	-	Read catalog, audit and the 3D city model (API, viewers)